

WESTPORT, New Zealand

WMO: 935150

Lat: 41.7406S

Lon: 171.5797E

Elev: 5

StdP: 101.27

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.9	2.9	-3.4	2.8	6.6	-1.7	3.3	6.2	14.9	11.6	12.8	11.9	3.4	160	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.9	22.3	18.1	21.4	17.7	20.5	17.3	19.4	21.0	18.7	20.3	18.1	19.7	4.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.6	20.2	18.0	13.0	19.6	17.3	12.4	18.9	55.3	21.0	53.0	20.3	51.0	19.8	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	10.8	9.4	0.5	24.2	0.8	1.4	-0.1	25.2	-0.5	26.0	-1.0	26.8	-1.6	27.8		
			-0.7	20.2	0.9	0.9	-1.4	20.8	-1.9	21.3	-2.4	21.8	-3.0	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	16.3	16.7	15.5	13.4	11.6	9.6	8.8	9.4	10.6	11.6	12.9	15.0
	DBStd	3.28	2.02	1.88	1.93	1.85	2.18	2.06	1.91	1.82	1.70	1.71	1.87	2.04
	HDD10.0	141	0	0	0	1	9	32	47	34	12	5	1	0
	HDD18.3	2110	67	52	89	148	209	263	294	278	231	209	164	106
	CDD10.0	1086	197	187	171	103	58	20	10	14	31	54	87	153
	CDD18.3	13	5	5	2	0	0	0	0	0	0	0	0	1
	CDH23.3	12	6	4	2	1	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.9	4.0	3.7	3.5	3.5	3.6	3.7	3.7	3.7	4.2	4.5	4.5	4.2
Precipitation	PrecAvg	1972	150	109	130	155	181	208	156	164	173	201	148	193
	PrecMax	2517	275	222	228	310	361	317	323	244	341	342	295	331
	PrecMin	1547	53	31	17	44	67	90	33	64	66	113	31	77
	PrecStd	234	51	44	58	77	70	64	70	46	71	64	80	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	23.7	22.4	19.8	18.5	15.7	14.9	15.7	16.8	17.7	20.2	22.2
		MCWB	19.5	18.8	17.2	16.0	14.3	12.7	11.1	11.2	12.6	13.9	16.0	17.6
	2%	DB	22.3	22.3	21.1	18.5	17.0	14.7	13.9	14.3	15.3	16.6	18.5	20.8
		MCWB	18.4	18.5	17.3	15.4	14.6	12.6	11.1	11.2	12.4	13.8	15.1	17.1
	5%	DB	21.2	21.4	20.2	17.8	16.1	14.0	13.2	13.5	14.6	15.7	17.4	19.6
		MCWB	17.7	18.0	17.0	15.2	14.0	12.2	10.8	11.0	12.1	13.2	14.5	16.4
	10%	DB	20.2	20.5	19.4	17.1	15.4	13.4	12.5	12.8	13.9	15.0	16.5	18.6
		MCWB	17.2	17.5	16.6	14.8	13.5	11.8	10.4	10.8	11.9	12.7	13.9	16.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.4	19.4	17.8	16.4	14.5	13.5	13.2	14.2	15.5	16.8	18.7
		MCDB	22.8	22.0	20.8	18.7	16.9	14.8	13.9	13.8	15.0	16.5	18.7	20.6
	2%	WB	19.3	19.5	18.6	16.8	15.5	13.6	12.5	12.5	13.6	14.6	15.9	17.8
		MCDB	21.0	21.2	20.0	17.5	16.2	14.2	13.2	13.3	14.4	15.6	17.5	19.3
	5%	WB	18.6	18.8	17.9	16.1	14.7	12.9	11.7	11.8	13.1	14.0	15.3	17.2
		MCDB	20.1	20.5	19.4	17.0	15.6	13.7	12.6	12.8	13.9	15.0	16.6	18.6
	10%	WB	17.8	18.1	17.2	15.3	13.9	12.1	10.7	11.2	12.5	13.4	14.6	16.7
		MCDB	19.6	20.0	18.8	16.5	15.1	13.0	12.1	12.4	13.5	14.5	16.0	18.1
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.9	6.9	6.8	6.6	6.4	6.8	6.6	6.3	6.2	6.3	6.0
		MCDBR	7.5	7.6	7.6	7.4	6.5	6.2	6.7	7.0	6.8	6.7	7.0	7.0
		MCWBR	4.7	4.7	4.8	5.1	4.8	4.8	4.9	4.9	4.9	4.8	4.6	4.5
	5% WB	MCDBR	6.0	6.2	6.0	5.3	5.1	4.8	5.4	5.6	5.3	5.2	5.4	5.3
		MCWBR	4.2	4.6	4.3	4.6	4.4	4.6	4.8	4.6	4.5	4.4	4.2	4.0
Clear-Sky Solar Irradiance	taub	0.353	0.343	0.326	0.314	0.302	0.299	0.291	0.306	0.334	0.343	0.348	0.345	
	taud	2.534	2.582	2.635	2.650	2.654	2.628	2.650	2.624	2.522	2.515	2.510	2.567	
	Ebn at Noon	974	958	932	876	820	784	825	872	906	947	972	988	
	Edh at Noon	108	98	85	72	61	57	59	72	93	103	110	105	
All-Sky Solar Radiation	RadAvg	6.30	5.62	4.27	2.81	1.82	1.41	1.67	2.38	3.43	4.60	5.87	6.21	
	RadStd	0.44	0.45	0.25	0.25	0.14	0.12	0.14	0.12	0.28	0.38	0.56	0.50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+4
(47)	Regional (4 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	-36	-90	+73

Nomenclature: See separate page